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13 June 1966

Superseding
XWS-4667(F)
14 November 1963

MILITARY SPECIFICATION

HONING MACHINE, DRY, PORTABLE

This Specification has been approved by the Naval Air Systems Command.

1. SCOPE

1.1 Scope - This specification covers one type of dry honing machine capable of removing corrosion products from aircraft and aircraft components.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

J-C-175	Cable Assembly, Power, Electrical (3-Wire, 3-Prong, grounding Plug Connector, for 125-Volt Equipment)
W-C-596	Connector, Plug, Electrical, Connector Receptacle, Electrical
QQ-A-225/8	Aluminum alloy, bar, rod, wire, and special shapes rolled, drawn or cold finished 6061.
QQ-A-250/11	Aluminum alloy 6061 - Plate and sheet
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements For
QQ-P-416	Plating Cadmium (Electrodeposited)
TT-C-490	Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings
TT-P-664	Primer, Coating Synthetic, Rust Inhibiting Lacquer-Resisting

FSC 4940

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SPECIFICATIONS

Federal (Cont'd)

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner

Military

MIL-P-116	Preservation, Methods of
MIL-C-744	Cleaning Abrasive, Spark Plug, Ceramic Insulator
MIL-E-7729	Enamel, Gloss
MIL-I-16910	Interference Measurement, Electromagnetic, Methods and Limits
MIL-G-21380	Grit, Abrasive, Blasting
MIL-A-22936	Abrasive, Glass Beads

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods
MS-33586	Metals, Definition of Dissimilar

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051) 5801 Tabor Road, Philadelphia, Pa. 19120.

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S Government Printing Office, Washington, D.C. 20402)

3. REQUIREMENTS

3.1 Preproduction Sample - The contractor shall furnish one dry honing machine to the Government for inspection and test in accordance with 4.3. The sample may be a preproduction model, or in cases where the contractor has a production model which meet the requirements of this specification, a sample may be taken from his stock. In either case, the sample shall be identical with the production models the contractor intends to supply. The approved preproduction sample shall be used as a reference standard for the inspection of the production models. Approval of the preproduction model does not relieve the contractor of his obligation to supply the production units in conformance with this specification. Fabrication and procurement of materials for any additional units or components thereof prior to approval of the preproduction model shall be at the contractor's risk and such units or components thereof will be subject to rejection in the event they fail to completely conform to this specification.

3.1.1 Standard Product - The dry honing machine delivered under this specification shall be the manufacturers standard commercial product, except for any design modification to meet requirements specified herein.

3.1.2 Standard Parts - Standard parts (MS, AN or JAN) shall be used wherever they are suitable for the purpose. Commercial utility parts such as screws; bolts, nuts, cotter pins, etc., may be used provided they posses

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suitable properties and can be replaced by the standard parts (MS, AN or JAN) without alteration; provided the corresponding part numbers are referenced in the parts list and on the applicable contractor's drawings. In the event there is no suitable corresponding standard part in effect on the date of invitation-for-bid, commercial parts may be used provided they conform to all the requirements of this specification.

3.2 Materials - Materials shall conform to the applicable specifications as specified herein. The material used in the construction of the equipment specified herein shall be new. Materials which are not covered by specification shall be of the best commercial quality and entirely suitable for the purpose intended.

3.2.1 Steel Components - All parts subject to abrasive impact wear, loads and fatigue shall be constructed of high alloy type steel, adequately heat treated or hardended to resist scratches, wear, deformation and fatigue imposed through normal service conditions.

3.2.2 Castings and Forgings - All castings shall be of uniform quality adequately braced, trussed or otherwise reinforced and completely free from blowholes, porosity, hand spots, shrinkage defects, cracks or other defects. Castings shall be free of flashings and sprues and shall be thoroughly cleaned of all scale, molding sand, and other extraneous materials. All exterior surface of castings shall be sanded or ground to provide a smooth, blended finish surface. All corners shall be chamfered, beveled, rounded, or otherwise blended to eliminate unnecessary service hazards to the operator. Forgings shall be free from scale, inclusions, mismatching and other defects which may affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein. In no event shall welding, brazing, soldering, peening or plugging, be used on castings or forgings for reclaiming any parts of the dry honing machine without authorization of the procuring activity.

3.2.2.1 Seasoning - All castings, after being cast and rough machined, shall be aged either naturally or artificially to insure dimensional stability.

3.2.3 Fungus-Proof Materials - Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermatically sealed, they shall be treated with a suitable fungicidal agent.

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3.2.4 Dissimilar Metals - Unless specified herein, dissimilar metals shall not be used in intimate contact with each other. Where practical dissimilar metals shall be protected against electrolytic corrosion. Dissimilar metals are defined in MS-33586.

3.3 Design and Construction - The dry honing machine shall be designed and constructed in such a manner as to provide a compact, self contained, light weight, portable, convenient and safe means of removing corrosion products from aircraft and aircraft components through the dry honing process. The machine shall be capable of being operated in the field without loss of abrasive material to the atmosphere. The basic machine shall be air operated and shall be complete with all necessary components, controls and attachments as hereinafter specified. When the machine is air and electrically operated an Electric Blower Conversion Kit shall be installed. The machine shall be of rugged and substantial design and shall be capable of continuous operation without excessive wear, fatigue, maintenance or repair. The machine and all components thereof shall be capable of withstanding a pressure test of ten inches of mercury vacuum without leakage or permanent deformation. The machine shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand strains, jars, vibrations and other conditions incident to shipping, storage and service.

3.3.1 Maintenance and Service - All working parts and components shall be designed and constructed to afford maximum operational performance with minimum maintenance. Adjustment shall be readily accomplished by operating personnel without special training, and without the need of special tools. All parts subject to wear, breakage or loss of adjustment shall be readily accessible for adjustment, repair or replacement.

3.3.2 Operation - The dry honing machine shall be capable of being operated by one operator. All controls such as valves, air pressure regulators, water separators and other components incident to the operation of the machine shall be located on the machine in such a manner as to afford optimum operation and utilization by one operator. All controls shall operate freely, with minimum friction, and without binding, sticking or jamming; however, the controls shall remain in selected positions during operation of the equipment.

3.3.3 Performance - The dry honing machine shall be capable of performing and repeat performance, with speed and accuracy, all types of dry honing operations utilizing all grades and types of abrasive material specified in MIL-C-744, MIL-G-21380, Type I and II and MIL-A-22936, in mesh sizes 70 to 500 as applicable. Adequate and dependable means of adjustment shall be designed and built into the machine to insure retention of uniformity of blast pressure and abrasive flow through its extended service life while being exposed to normal dry honing operations.

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3.3.4 Fastening Devices - All screws, pins, bolts and similar parts shall be installed with adequate means to prevent loss of proper torque and adjustment. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.5 Threads - All threaded components shall be designed in accordance with the applicable requirements of the National Bureau of Standards Handbook H28.

3.3.6 Safety Devices - Each machine shall be equipped with safety devices to adequately protect the operator from all safety hazards incident to its operation.

3.3.7 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. All replaceable parts, where practicable shall be permanently and legibly marked with the manufacturer's part number.

3.3.8 Dimensions and Weight - Dimensions and weight of the dry honing machine shall not exceed the following:

Height	36 inches
Width	21 inches
Width Including Hose	24 inches
Length	31 inches
Weight (Air Version)	160 pounds
Weight (Electric Blower Version)	188 pounds

3.4 Components - The dry honing machine shall consist of the following principal components mounted on a two wheel carriage assembly.

(a) Dust Collector
(b) Abrasive Reclaiming System
(c) Abrasive Hopper
(d) Abrasive Supply and Return Hoses
(e) Blast Gun Assembly
(f) Pneumatic System
(g) Necessary Control and Regulator Valves and Gages
(h) Accessories
(i) Auxiliary Electric Driven Vacuum Pump (When Required)

3.4.1 Carriage Assembly - The dry blast machine shall be mounted on a carriage assembly having two supporting legs, wheels, floor contacting feet and a handle.

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3.4.1.1 Legs - The supporting legs shall be constructed of a minimum of 3/4 inch diameter, one piece steel or aluminum solid rod. The legs shall be reinforced as necessary to carry the intended loads and weight of the equipment without buckling, swaying or sagging. The legs shall be recessed flush into the sides of the dust collector box and shall be bolted into position. The legs shall be equipped with two wheels on the rear end to provide complete portability and two support feet on the front end to provide complete operating stability.

3.4.1.2 Wheels - The wheels shall be six inches in diameter with a minimum tread width of two inches and shall be equipped with roller bearings or equivalent. The face of the wheels shall be of 80 shore durometer hardness buna SBR or equal construction. The wheels shall be mounted inboard of the legs and shall be supported by one continuous axle between the legs. The wheels shall be mounted on the axle in such a manner as to prevent shifting of position.

3.4.1.3 Feet - The floor contacting feet shall be soft molded rubber no less than 2 inches in diameter to insure maximum stability on a metallic deck surface. The rubber shall be vulcanized to a steel plate of at least 1/8 inch thick and attached to the legs with a threaded stud.

3.4.1.4 Handle Assembly - The handle assembly shall be constructed of 3/4 inch formed steel tubing. The handle shall be formed in a manner that will place the end of the handle from 30 to 32 inches above the floor when the machine is balanced on the two wheels in the normal transporting position. The handle shall be securely mounted in the center of the dry honing machine and shall include a nonslip handle bar grip for ease of transportation.

3.4.2 Dust Collector - The dust collector shall consist of a rectangular, double wall, bag house, fabricated of not less than 14-gauge (0.064 inch) aluminum alloy conforming to QQ-A-250/11 with adequate stiffners on the front and back. Filter elements of the dust collector shall be of the cloth multiple envelope type, each envelope being independently replaceable but shall be mounted as a set in a rigid frame, such that the complete set may be lifted free of the bag house enclosure for periodic maintenance. A suitable seal of felt or rubber shall be provided to seal the individual envelopes to the frame, and to seal the complete frame assembly to the bag house enclosure. The assembly shall be secured by clamps without recourse to threaded fasteners or other devices sensitive to abrasive jamming. A manually operated shaker mechanism shall be provided to permit shaking the bag filters to remove dust. The removed dust

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shall be collected on a tray at the bottom of the bag house. Means shall be provided for easy removal of the dust collector tray. The top panel or corner of the bag house shall be readily removable in order that the filter bags may be removed. The dust collector shall be rated at three inches water gage pressure drop at 100 c.f.m. flow. Collapsible handles shall be included on each side of the dust collector box for lifting convenience.

3.4.3 Abrasive Reclaimer - The abrasive reclaimer shall receive all abrasive returned by the vacuum system after blasting against the work piece and shall separate the abrasive particles from the conveying air stream, depositing reusable abrasive on a vibrating screen and carrying the unusable dust into the dust collector box. The abrasive reclaimer shall be capable of satisfactory completing all performance tests specified in Section 4. The abrasive inlet shall be so oriented as to impose minimum flexure of the vacuum return hose when stowed in the carriage position. Hose in the stored position shall not extend more than three inches beyond the side of the machine.

3.4.4 Auxiliary Electrically Driven Vacuum Pump - The vacuum system shall be so designed as to provide a simple conversion to include an auxiliary electric driven vacuum pump to assist when the available air supply is insufficient to fully actuate the ejector.

3.4.4.1 Electrically Driven Vacuum Pump Conversion Kit -When specified on electrically driven vacuum pump conversion kit shall be supplied as a separate item or included on the basic machine (See 6.2). The conversion kit shall include an electrically driven vacuum pump and all brackets, piping and electric cord to attach the pump to the dry honing machine.

3.4.4.1.1 Electrically Driven Vacuum Pump - The pump shall be of such design and size as to render the dry honing machine operable with a total compressed air consumption of not more than 60 cubic feet per minute (CFM) when it is installed. The pump shall be explosion proof for class 1, group D or class II, group E, F, or G use. The motor shall be at least 7/8 horsepower, and rated for continuous duty at 75°F. The electric motor shall be radio interference suppressed in accordance with the requirements of MIL-I-16910. The pump shall not draw more than 10 amps. It shall be fitted with an explosion proof starter switch and equipped with a suitable overload cutout. The weight of the pump, including the switch shall not exceed 22-1/2 pounds.

3.4.5 Abrasive Hopper and Feed System - A conical shaped hopper having a minimum abrasive capacity of five pounds of aluminum oxide shall be provided. The hopper shall continuously receive abrasive as discharged by the abrasive reclaimer and shall supply reusable abrasive to the feed system. The hopper shall be a seamless steel spinning of at least 14 gauge thickness and

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shall incorporate quick release clamps for ease of removal. A suitable feed system not employing moving parts shall be provided to assure uniform delivery of all abrasives specified in 3.3.3 and provide automatic purging of the abrasive supply hose. A handle shall be incorporated on the hopper to facilitate handling. A visual means shall be provided to return the abrasive feed back to the abrasive hopper when the operator ceases blastings.

3.4.6 Vibrating Screen with Magnetic Separator - A vibrating screen shall be mounted between the abrasive reclaimer and abrasive hopper to entrap oversize debris particles passed by the abrasive reclaimer. The screen shall be 14 mesh size, constructed with .023 inch diameter steel wire, or equivalent aperature perforated steel plate. The screen shall be vibrated to a present amplitude by an air driven vibrator. The vibration amplitude shall be sufficient to prevent abrasive build-up on the screen. A permanent magnet, capable of separating and retaining magnetic particles shall be provided. The magnet shall be mounted directly in the path of all abrasive materials being returned to the abrasive hopper in such a manner as to insure maximum retention of magnetic particles. The screen shall be provided with adequate edge gaskets to insure that all materials pass through the screen rather than bypass it.

3.4.7 Blast Gun Assembly - The blast gun assembly shall consist of a hand held gun, control valve, nozzle assembly, abrasive supply hose, abrasive return hose and air hose.

3.4.7.1 Hand-Held Gun - The hand-held gun shall be of cast aluminum construction, approximately 7 1/2 inches overall height and shall be designed to be convenient to operate by one hand and not cause undue fatigue to the operator. The lower internal end of the gun shall be protected by a replaceable rubber liner having a minimum length of 1 1/4 inches. The nozzle end of the gun housing shall be constructed in such a manner as to allow easy snap-on mounting of brushes, pads and accessories listed in 3.4.11. When moving over a flat surface at a rate of one lineal foot in ten seconds and blasting at 80 psi with 320 mesh aluminum oxide, the gun shall deliver an abrasive flow of 2.0 pounds per minute and produce a cleaned path not less than one inch wide.

3.4.7.2 Control Valve - A control valve shall be mounted on the blast gun housing to control both the dry honing (blast) operation and the amount of vacuum created on the abrasive supply hose. The control valve shall be equipped with a palm-actuated lever not less than 3 inches in length. The valve shall be spring loaded in the normally closed position.

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3.4.7.3 Nozzle Assembly - The nozzle assembly shall consist of tungsten carbide nozzle with an air jet mounted in a machined aluminum body housing. The nozzle shall be designed to provide the delivery and cleaning path specified in 3.4.6.1. The nozzle assembly shall be designed and constructed in such a manner that it can be readily replaced.

3.4.7.4 Abrasive Supply Hose - The abrasive supply hose shall be 5/8 inch I.D. with 1/8 inch wall thickness, and shall be clear, transparent plastic to permit direct observation of the abrasive flow. The plastic composition shall be extremely flexible, shall have a tensile strength of not less than 1450 psi, shall have a durometer hardness of 60-64 shore A, and shall retain its flexibility and shall be suitable for operation over a temperature range of -20 degrees to + 165 degrees Farhenheit. The composition shall be rated for an abrasion resistance by the Taber method of not more than 4.0 milligrams per 1,000 cycles. The abrasive supply hose shall be at least twenty (20) feet in length.

3.4.7.5 Vacuum Return Hose - The vacuum return hose shall be of sufficient size to return the spent abrasive to the abrasive reclaimer. The hose shall be lined with a minimum of 1/32 inch static conducting natural rubber and shall be suitable reinforced and covered so as to resist a 10 inch mercury internal vacuum. The vacuum return hose shall be at least 20 feet long and shall not weigh more than 10 pounds.

3.4.7.6 Air Hose - A suitable rubber hose, of 3/8 inch bore, shall be provided to conduct air to the gun housing. The hose shall not be less than 20 feet long.

3.4.8 Pneumatic System - The machine shall be capable of satisfactory operation on air pressures ranging from 80 to 100 psi and air flows ranging from 80 to 90 cfm. All components in the system shall be capable of withstanding pressures up to 150 psi without damage. The pneumatic system shall consist of an air ejector pump or electric vacuum pump, when specified, pressure regulator, screen vibrator (see 3.4.5), air filter, valving and pressure gage.

3.4.9 Pressure Regulator and Gauge - A 150 pound maximum variable pressure regulator with gauge shall be incorporated on the air line supplying the blast gun assembly and shall permit regulated control of the abrasive working pressures in order that variable surface finishes may be produced when used with different mesh size abrasives. The pressure regulator shall be mounted on the back of the dust collector box in an area readily accessible to the operator.

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3.4.10 Air Ejector Pump - An air ejector pump constructed entirely of aluminum alloy capable of producing a minimum vacuum at the inlet of 40 inch WG (water gage) static pressure at 100 cfm induced flow, with a supply pressure of 80 psig shall be provided. The air ejector pump shall incorporate a quick disconnect on the supply hose for cleaning. The ejector pump shall be constructed in such a manner so that maximum efficiency can be maintained throughout its service life without the need of excessive maintenance. The unit shall be designed and constructed in such a manner that maintenance can be performed with a minimum of disassembly. A suitable muffler shall be fitted at the outlet of the ejector pump to reduce the overall operational noise level.

3.4.11 Air Filter - An air filter incorporating a moisture separator and screen shall be installed in the air supply line and mounted on the machine ahead of all air pressure operated components. The air filter shall be capable of removing at least 95% of the condensate and other contaminates from the air supply line which may be harmful to the air operated components. The filter shall incorporate a clear shatter resistant platic globe and drain valve assembly. The plastic globe shall be capable of being squeezed three-quarters of its diameter without evidence of cracks or breaks.

3.4.12 Coupler valve and pressure gauge - A 1/2-inch I.D. quick disconnect coupler and integral three-way valve assembly shall be furnished to facilitate hook-up operations with any fleet operation air supply. The coupler valve shall be furnished as a complete unit and shall include both the male adapter and the female three-way valve type coupler. The three-way valve coupler shall employ an "O"-ring type sliding shut-off valve assembly. The coupler shall utilize a positive coupling lock, and the operator shall not be required to work against air pressure during either coupling or uncoupling operations. A pressure gauge shall be installed downstream from the coupler valve and filter assembly to indicate supply line pressure. The minimum operational supply line air pressure of 80 p.s.i. shall be indicated by a red line on the gauge.

3.4.13 Hose Rack and Storage Compartment - A hose rack and storage compartment shall be mounted on the foot of the machine to permit storage and transportation of all necessary air, abrasive and vacuum hoses as well as the necessary brushes and accessories. The hose rack shall be designed to retain the hose while the machine is being transported to and from the job. The storage compartment shall be at least 6 x 10 x 12 inches and shall be mounted on the front of the machine within the normal confines of the hose when in the coiled transportation position. The compartment shall be manufactured of not less than 14 gage aluminum alloy construction and shall be provided with a hinged door and latch assembly capable of preventing accidental opening during transportation or normal operating conditions.

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3.4.14 Accessories and Spare Parts - When specified in addition to the basic machine, the following accessories and spare parts shall be provided: (see 6.2).

(a)	Six	(6)	flat surface brushes
(b)	Two	(2)	inside corner brushes
(c)	Two	(2)	outside corner brushes
(d)	Two	(2)	uneven surface cleaners
(e)	One	(1)	spare blast nozzle assembly
(f)	Six	(6)	brush retainers

3.4.14.1 Brushes - All brushes and accessories shall be manufactured to definite standards, tolerances and clearances in order that they can be readily attached to the blast gun assembly. All brushes shall be constructed of 0.014 inch diameter crimped nylon bristles, and shall be designed and manufactured in such a manner as to retain the abrasive without spilling to the atmosphere. All brushes shall be at least 1-1/2 inch in diameter.

3.4.14.1.1 Flat Surface Brushes - Flat surface brushes shall have a one inch trim length.

3.4.14.1.2 Inside Corner Brushes - Inside corner brushes shall be 1-3/4 inch trim length. The brush shall be trimmed in a manner to fit a 90° inside corner.

3.4.14.1.3 Outside Corner Brush - The outside corner brush shall be two inch trim length and shall be stiffened by an outside corner adapter. (See 3.4.12.3.1). The brush shall be trimmed to fit a 90° outside corner.

3.4.14.1.3.1 Outside Corner Adapter - The outside corner brush nozzle adapter shall be constructed of welded or machined aluminum. The adapter shall be constructed in a manner to affort adequate support to the bristles and thereby minimize the loss of abrasive materials when operating on difficult to clean outside corners. The nozzle adapter shall fit snugly around the nozzle of the blast gun and shall be held in place by the thumb screw. Knock out holes shall be furnished to facilitate brush removal.

3.4.14.1.4 Uneven Surface Cleaner - The uneven surface cleaner shall consist of a six-inch diameter doughnut of three-inch thick, open cell polyurethane foam. The center hole shall be 1-1/2 inches in diameter. The polyurethane sponge material shall be of two pounds per cubic foot density. The sponge material shall be fastened to a stiff metal aluminum plate 1/8 inch in thickness which shall be welded to a nozzle adapter. The nozzle adapter shall readily attach to the blast gun assembly.

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3.4.15 Maintenance and Operational Instruction Plate - A photo engraved metal maintenance and operational instruction plate shall be permanently attached to the top cover of the machine. The plate shall be a maximum of 6 x 18 inches and shall be permanently and legibly marked with daily maintenance instructions and shall include brief operational instructions. The plate shall include under operational instructions the nominal pressures to be utilized with various mesh sizes and types of abrasives. This information will be furnished to the contractor by the Commanding Officer, Naval Air Station, North Island, California, after award of contract.

3.4.16 Identification Plate - An identification plate permanently and legibly marked in accordance with the requirements of MIL-STD-130 shall be permanently attached to each dry honing machine.

3.5 Finish - All molded, welded or fabricated sheet metal and structural steel parts, castings and forgings of the machine shall be clean free from sand, dirt, fins, sprues, scale, flux and other harmful extraneous material. External surfaces shall have a smooth finish free from burrs, flaws, ragged and sharp edges.

3.5.1 Ferrous Materials - Ferrous surfaces shall be given protective treatments in either of the methods indicated below. All ferrous items such as screws, bolts, nuts, washers, etc., shall be cadmium plated in accordance with QQ-P-416, Type II or III as applicable, or zinc plated in accordance with QQ-Z-325. Ferrous structural surfaces shall be dry blasted followed by the protective Paint System Specified in 3.5.2.

3.5.2 Painting - All metal surfaces of the dry blast machine on which a protective coating will not be detrimental to proper functioning shall be painted as follows:

Apply a pre-treatment using
TT-C-490 Type I (Phosphate
Type) followed by 1 coat of
TT-P-664 primer coating,
synthetic, rust inhibiting, lac-
quer resisting, and 2 coats
of MIL-E-7729 gloss enamel.
Use yellow color number 13538
of FED-STD-595

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3.6 Workmanship - The honing machine, including all parts and accessories, shall be constructed and finished free from all defects that would affect the proper functioning in service. Particular attention shall be given to neatness and thoroughness of, welding, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. The honing machine shall be thoroughly cleaned of loose, spattered or excess rosin flash, metal chips and other foreign matter, during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection - The inspection and testing of the portable dry honing machine shall be classified as follows:

(a) Preproduction inspection
(b) Quality conformance inspection

4.3 Preproduction inspection - Preproduction inspection shall be applied to the preproduction sample submitted in accordance with 3.1. Failure of the preproduction sample to pass examination or one or more preproduction tests shall be cause for rejection.

4.3.1 Unless otherwise specified (see 6.2), preproduction inspection shall consist of the examination and all tests in 4.5 and 4.6.

4.4 Quality Conformance Inspection - Quality conformance inspection shall consist of the following:

(a) Individual Inspection (4.4.1)
(b) Sampling Inspection (4.4.2)

4.4.1 Individual Inspection: Each honing machine shall be examined in accordance with 4.5.

4.4.2 Sampling Inspection: Samples shall be selected at random in accordance with MIL-STD-105 inspection level S-2 acceptable quality level (AQL) 2.5 and subject to the following tests:

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(a) Abrasive Capacity Test 4.6.1
(b) Over Pressure Test 4.6.3
(c) Vacuum Pressure Test 4.6.4
(d) Purge Test 4.6.5.7

4.5 Examination - The portable dry honing machine and components shall be thoroughly examined for design, construction, materials and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.6 Test Methods -

4.6.1 Abrasive Capacity Test - The dry honing machine shall be operated using the smallest and largest available mesh size between 70 and 500 of abrasives specified in 3.3.3. Failure of the machine to operate satisfactorily using all these abrasives will be cause for rejection.

4.6.2 Air Capacity and Pressure Test - The dry honing machine shall be operated on 80 psi, 80 cfm of air and at 100 psi, 90 cfm of air, and at 60 cfm of air with the auxiliary electrically drive pump installed (see 3.4.4). Failure of the machine to operate satisfactorily under these conditions will be cause for rejection.

4.6.3 Over Pressure Test -The machine shall be subjected to air pressure of 150 psi. Failure of any component of the machine shall be cause for rejection.

4.6.4 Vacuum Pressure Test - The complete dry honing machine shall be subjected to a pressure test of ten inches of mercury vacuum for a period of not less than five minutes. Failure, permanent deformation and leakage of any component or part of the machine shall be cause for rejection.

4.6.5 Performance Tests - The dry honing machine shall be subjected to the performance tests hereinafter specified. Performance tests shall be conducted with the blast gun assembly moved over the surface at the rate of one foot in ten seconds. Aluminum oxide 320 mesh size abrasive and 80 psi air pressure shall be utilized for all performance tests.

4.6.5.1 Spray Pattern Test - While blasting under conditions specified in 4.6.5 move the blast gun over a flat surface in a straight path, for a distance of three feet. If the cleaned path is less than one inch at any place along the length of the path the machine shall be rejected.

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4.6.5.2 Flat Surface Test - While blasting under conditions specified in 4.6.5 using the flat surface brush, blast a one square foot area of flat metal plate in the down-hand, vertical and overhead positions. Escape of any abrasive or failure to blast in a uniform manner will be cause for rejection.

4.6.5.3 Curved Surface Test - While blasting under conditions specified in 4.6.5 blast a one square foot area of a 4-inch O.D. metal pipe in the down-hand, vertical and overhead positions. Escape of any abrasive or failure to blast in a uniform manner will be cause for rejection.

4.6.5.4 Outside Corner Test - While blasting under conditions specified in 4.6.5 using outside corner brush and outside corner adapter blast a 90° metal outside corner surface for a length of 3 feet. Excessive loss of abrasive or failure to blast in a uniform manner will be cause for rejection.

4.6.5.5 Inside Corner Test - While blasting under conditions specified in 4.6.5 using the inside corner brush blast a 90° metal inside corner for a length of 3 feet. Excessive loss of abrasive or failure to blast in a uniform manner will be cause for rejection.

4.6.5.6 Uneven Surface Test - While blasting under conditions specified in 4.6.5 using the uneven surface cleaner blast a two square foot area of metal plate having compound convex and concave curves of not more than 12 inch radii. Escape of any abrasive or failure to blast in a uniform manner will be cause for rejection.

4.6.5.7 Purge Test - While blasting under conditions specified in 4.6.5 release the blast control valve to stop the blast action and visually observe the abrasive flow in the transparent abrasive supply hose. Failure of the system to reverse direction of abrasive flow in the hose so as to completely purge the hose within 10 seconds (or 15 seconds when using the auxiliary electrically driven pump) shall be cause for rejection.

4.6.5.8 Dust Test - While blasting under conditions specified in 4.6.5 observe that no visible dust is discharged from any part of the machine or lines, with particular emphasis on the air ejector outlet. Evidence of dust will be cause for rejection.

4.6.5.9 Blast Capacity Test - While blasting under conditions specified in 4.6.5 and with a five-pound charge in the abrasive hopper blast against a steel plate for a period not less than 30 minutes. Failure to deliver a uniform blast throughout the test period will be cause for rejection.

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4.6.5.10 Abrasive Flow Test - While blasting under conditions specified in 4.6.5 and with a five (5) pound charge of abrasive in the hopper, raise the gun clean of the work surface and blast for one minute, allowing the discharge abrasive to escape from the system. Weigh the remaining abrasive in the hopper. The presence of more than three (3) pounds and less than two and three quarter, (2 3/4) pounds of abrasive in the hopper at the termination of the test will be cause for rejection.

4.6.5.11 Air Filter Globe Test - The air filter globe shall be placed in a vise and squeezed to one-quarter of its original diameter. Any evidence of cracks or breaks shall be cause for rejection.

4.6.6 Radio Interference Suppression Test: When provided the electric motor on the auxiliary electrically driven vacuum pump shall be tested for conformance to MIL-I-16910.

4.7 Examination of Preservation, Packaging, Packing and Marking for Shipment and Storage - Preservation, packaging, packing and marking for shipment and storage shall be examined to determine conformance with the documents referenced in Section 5.

4.8 Rejection - Rejected equipment shall not be retested until the extent and cause of failure have been determined and appropriately corrected. The contractor shall record the cause of failure and the action taken to preclude recurrence. After correction, all of the test shall be completed.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - All dry honing machines shall be cleaned in accordance with procedure D-1 or D-2 and preserved and packaged in accordance with Method III of Specification MIL-P-116.

5.2 Packing -

5.2.1 Overseas Shipment - All components comprising one complete dry honing machine shall be packed in a box conforming to PPP-B-601 (overseas type), or PPP-B-621 (Class 2). Blocking, bracing and anchoring shall be in accordance with MIL-STD-1186. Boxes shall be closed and strapped in accordance with the applicable box specification or the appendix thereto. Flat steel strapping shall be type I, Class B.

5.2.2 Domestic Shipment - All components comprising one complete dry honing machine shall be packed in a shipping container in such a manner that will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at the lowest rate.

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5.3 Marking - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended Use - The portable dry honing machine is a compact, self-contained, lightweight, portable unit used for honing small work pieces and for the safe and convenient removal of corrosion products from aircraft and aircraft components through the dry honing process. The machine is air-operated and can be used in shore-based or shipboard operations without loss of abrasive material to the atmosphere.

6.2 Ordering Data - Procurement Documents shall specify the following:

(a) Title, Number and date of this specification.
(b) Accessory equipment when required (see 3.4.14).
(c) Finish if different (see 3.5).
(d) Selection of the applicable levels of preservation and packaging (see section 5).
(e) Whether the auxiliary electrically driven vacuum pump is required (see 3.4.4.1).

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Naval Air Systems Command
Washington, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-H-81324(AS) HONING MACHINE, DRY, PORTABLE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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